

# Snehil Sundriyal

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## Education

**BITS Pilani — M.Sc. Physics & B.E. Electrical and Electronics Engineering**

*Oct 2022 – Aug 2027*

Relevant Coursework: Computer Programming, Internet of Things, Digital Design, Microprocessors, Probability & Statistics

## Skills

**Languages:** C/C++, Go, JavaScript, Python, SQL

**Frameworks & Technologies:** NodeJS, ExpressJS, ReactJS, MongoDB, Docker, Microservices, Playwright

**AI & Automation:** OpenAI APIs, LangChainGo, Workflow Automation, Prompt Engineering, AI-Assisted Development

**Tools & Platforms:** Git, Gallabox, Postman

## Internships

**Ripplika (Remote)**

*April 2026 – Present*

*Software Engineering Intern (AI Automation)*

- Built and deployed **4+ production workflows** using **Playwright automation**, **AI agent nodes**, and workflow functions.
- Designed scalable document verification pipelines with **automated decision flows** and asynchronous task execution.
- Integrated workflow infrastructure in client systems, enabling **100+ daily sessions** with OTP handling and live tracking.
- Leveraged **Cursor and AI-assisted workflows** to accelerate prototyping, debugging, and production implementation.
- Collaborated with clients to debug integrations, optimize **UX flows**, and resolve **20+ workflow edge case failures**.

**CSIR-CSIO, Chandigarh**

*May 2024 – Aug 2024*

*Research and Development Intern*

- Simulated and optimized **15+ optoelectronic devices**, improving **coupling efficiency** and propagation via modeling.
- Modeled **10+ circuit architectures** using **INTERCONNECT** and **FDTD**, enabling accurate time-domain analysis.
- Performed **modal analysis** on **8+ waveguide structures**, identifying performance trade-offs in optical systems.
- Executed **50+ simulation iterations**, systematically refining parameters to achieve **operational metrics**.
- Characterized performance trends across **20+ optical simulation datasets** for advanced **photonic analysis studies**.

## Projects

**AI Workflow Orchestration System for Udyam Registration**

- Built a **120+ node automation workflow** for document processing and **real-time user interaction management**.
- Engineered structured validation pipelines for Aadhaar, PAN, and banking verification using **modular workflow schemas**.
- Integrated **Gallabox chat infrastructure** for **OTP handling**, live tracking, and automated user communication flows.

**AI Itinerary Planner — GenAI Application**

- Built a **Go-based backend service** that generates personalized itineraries from **user preferences** and constraints.
- Integrated **LLM-powered itinerary generation** using **LangChainGo** and **OpenAI API** for dynamic recommendations.
- Architected **asynchronous workflows** with **UUID-based tracking** for scalable AI request handling in production.

**Location-Verified Attendance System — Data & Web Systems**

- Built a **location-verified attendance system** supporting **350+ users/session** across live academic lecture workflows.
- Implemented **geolocation validation** and automated attendance capture across **300+ records/session** in production.
- Deployed the system for lecture attendance, reducing **manual processing time** by **70%** across departments.

**Go Movies — Backend & Full-Stack Development**

- Developed a scalable platform with **20+ REST APIs** and **5+ relational entities** for efficient data management.
- Implemented **JWT-based authentication** and **role-based access control** across authenticated user workflows.
- Built and integrated a **React frontend** supporting CRUD operations, filtering, and pagination for administrators.

## Leadership & Activities

**Core Member, Music Club — BITS Pilani**

*Jul 2023 – Aug 2027*

- Coordinated **large-scale events** with **300+ participants**, managing logistics, execution, and cross-team collaboration.
- Performed as a **vocalist** at flagship institute cultural festivals including Oasis and APOGEE inaugurations.